Biological Insights from Genome-Wide Association Studies and Whole Genome Sequencing of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

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Research Article

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Article

Biological Insights from Genome-Wide Association Studies and Whole Genome Sequencing of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

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Abstract

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating disorder of poorly understood etiology. We performed a meta-analysis of two European-ancestry ME/CFS genome-wide association studies (GWAS) with no overlap in subjects, DecodeME and Million Veteran Program, comprising a total of 19,470 cases and 699,111 controls. Post-GWAS analyses investigated the association between ME/CFS and specific tissues, cell types, cellular components, and canonical pathways. Findings were independently evaluated for replication against a module of ME/CFS risk genes previously prioritized by machine learning applied to rare coding variants from whole-genome sequencing (WGS) of ME/CFS cases and controls of European ancestry. Tissue enrichment analysis implicated multiple brain regions and the pituitary, with no peripheral tissue reaching significance, and three survived Bonferroni-corrected replication. Gene-set analysis identified multiple neuronal and synaptic gene sets, several of which were independently replicated, with glutamatergic synapses as the most specific replicated signal. Cell-type analysis identified independent replicated signals in distinct neuronal populations of subcortical and cerebellar regions. These results suggest a role for synaptic function in specific brain regions in the pathogenesis of ME/CFS, with convergent support from both common and rare variant data. Larger studies are needed to confirm these findings and to identify specific targets for therapeutic intervention.

Keywords: ME/CFS; myalgic encephalomyelitis; genome-wide association study; meta-analysis; METAL; MAGMA; synapse; post-GWAS

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1. Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating disorder of unknown etiology, currently without FDA-approved treatments [1]. It is defined by persistent fatigue, post-exertional malaise, cognitive impairment, and orthostatic intolerance lasting at least six months [2, 3]. A meta-analysis on 44 datasets reported a pooled prevalence of 0.65% (95% CI 0.43–0.99) in the adult population, with a female-to-male ratio of approximately 1.5 to 2 [4]. Incidence shows two peaks, one in adolescence and one in the fourth decade of life [5], with early-onset disease associated with greater disease severity, higher rates of affected first-degree relatives, and more frequent infectious triggers [6]. Remission is uncommon and the majority of patients experience persistent functional impairment, with most not returning to their pre-illness occupational status [7]. Cognitive impairment is one of the most commonly reported symptoms, with deficits in attention, memory, and processing speed [8]. Neuroimaging studies consistently document additional brain area recruitment during cognitive tasks [9], suggesting compensatory neural mechanisms underlying these deficits. In the DecodeME cohort, cognitive symptoms were more prevalent than disabling fatigue, with 94.5% of cases reporting confusion or "brain fog" [10].

An interplay between genetic predisposition and environmental triggers is suggested by an increased prevalence of the disease among both relatives and spouses/partners of affected individuals [11–13]. Post-infectious onset has been documented after a heterogeneous range of pathogens, including Epstein-Barr virus, *Coxiella burnetii*, Ross River virus, and SARS-CoV-2 [14–16].

The first large-scale GWAS of ME/CFS, conducted by the DecodeME consortium on 15,579 clinician-confirmed cases, identified eight risk loci mapped to genes involved in innate immunity and brain function [10]. At present, summary statistics from three other cohorts are publicly available, as shown in Table 1, where a Long COVID GWAS was also included, given an ME/CFS prevalence of approximately 51% among patients with Long COVID [16]. The Million Veteran Program (MVP), with 3,891 cases derived from electronic health records (EHR), is the largest after DecodeME, while the UK Biobank (UKB) and FinnGen cohorts have smaller case counts. Besides DecodeME, only the UKB Neale Lab female subset reached genome-wide significance.

Additional insights on the genetic basis of ME/CFS may emerge from rare variants and monogenic presentations, as already documented in other common diseases [17,18]. Whole-genome sequencing (WGS) in 20 severely ill ME/CFS patients identified an increased burden of likely pathogenic and pathogenic variants in nine genes, most of which have been implicated in cognitive symptoms and neurological function [19]. Another WGS study of 283 ME/CFS patients and 213 controls of European ancestry identified 115 genes associated with ME/CFS based on prioritization of rare coding variants by machine learning, with enrichment for immune and neurological pathways [20]. Rare monogenic presentations have also been documented, with one case implicating a structural variant at the *AKR1C* locus leading to increased synthesis of inhibitory GABAergic neurosteroids [21].

While meta-analysis of summary statistics is as effective as pooling individual-level data [22], this avenue is constrained here by substantial overlap in controls between DecodeME and the UK Biobank, and between Long COVID and both FinnGen and UK Biobank [23]. The largest meta-analysis with no overlap in controls is therefore the one combining DecodeME and MVP, which we present here. The resulting summary statistics were used as input for three downstream analyses: gene-tissue expression, cell-type expression at single-cell resolution, and gene-set enrichment; all implemented within the FUMA pipeline [24–26]. As an independent replication resource, we use the 115 ME/CFS risk genes prioritized by HEAL2, a new machine learning algorithm trained on rare coding variants [20].

Table 1. Publicly available ME/CFS GWAS summary statistics.

Cohort	Cases	Controls	N_{eff}	GWS	Trait	Ref.
DME-1	15,579	259,909	58,792	250	ME/CFS (CCC/IOM)	[10]
DME-2	15,579	155,790	56,651	198	ME/CFS (CCC/IOM)	[10]
DME-1 F	12,833	218,949	48,490	211	ME/CFS (CCC/IOM)	[10]
DME-1 M	2,746	40,960	10,294	0	ME/CFS (CCC/IOM)	[10]
DME-1 PI	9,738	259,909	37,545	3	PI ME/CFS (CCC/IOM)	[10]
MVP	3,891	439,202	15,427	0	PheCode 798.1 CFS	[27]
UKB-EIB	2,092	482,506	8,332	0	Self-reported CFS	[28]
UKB-NL	1,659	359,482	6,606	0	Self-reported CFS	[29]
UKB-NL F	1,208	192,945	4,802	11	Self-reported CFS	[29]
UKB-NL M	451	166,537	1,799	0	Self-reported CFS	[29]
FinnGen	283	463,029	1,131	0	Post-viral fatigue	[30]
Long COVID	6,450	1,093,995	25,649	0	Long COVID	[23]

Assembly is GRCh38 for DecodeME (DME), Million Veteran Program (MVP), FinnGen, and Long COVID; GRCh37 for UK Biobank cohorts. All cohorts are of European ancestry except FinnGen (Finnish), but for MVP and Long COVID multiancestry summary statistics are also available. Regressions are logistic, except the summary statistics generated by Neale Lab (NL) and the European Institute of Bioinformatics (EIB). N_{eff}: effective sample size ($4/(1/N_{\mathrm{cas}} + 1/N_{\mathrm{con}})$). Only DME and NL provide sex-specific summary statistics. GWS: number of genome-wide significant variants ($P < 5 \times 10^{-8}$) after quality control (INFO > 0.9, MAF > 0.01), lift-over to GRCh37 (if applicable), and harmonization with MungeSumstats [31]. The present study uses DME-1 and MVP. CCC: Canadian Consensus Criteria; IOM: Institute of Medicine criteria. F: female; M: male; PI: self reported post-infectious onset.

2. Materials and Methods

Summary statistics from two publicly available ME/CFS GWAS (DME-1 and MVP, Table 1) were quality-controlled, harmonized, and combined by meta-analysis. The resulting meta-analytic summary statistics were then used as input for gene-set, tissue expression, and cell-type expression analyses using MAGMA within the FUMA SNP2GENE and Cell Type modules [24–26], each evaluated for replication using an independent ME/CFS rare-variant gene module [20]. An overview of the pipeline is provided in Figure 1.

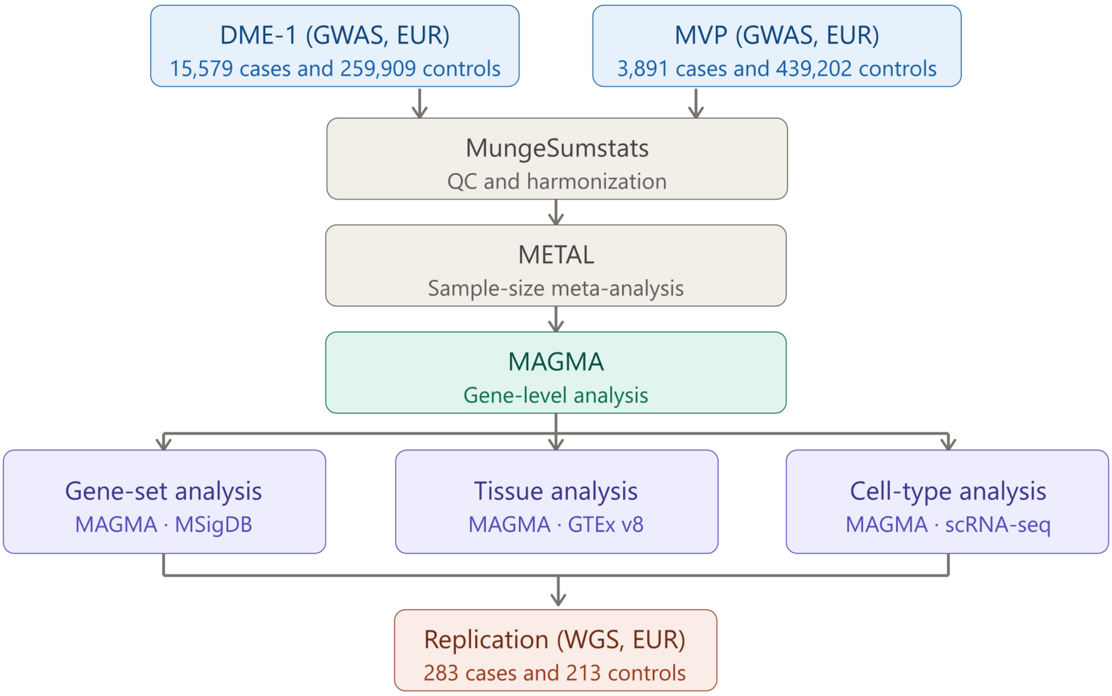


Figure 1. Analysis pipeline. Summary statistics from two ME/CFS genome-wide association studies (DecodeME, DME-1, $n = 15{,}579$ cases of European ancestry; and the Million Veteran Program, MVP, $n = 3{,}891$ cases of European ancestry) were quality-controlled and harmonized using `MungeSumstats`, then combined by sample-size-weighted Z-score meta-analysis in METAL. Gene-level association statistics were computed with MAGMA and used as input for three parallel downstream analyses: gene-set analysis against MSigDB 2023.1.Hs, gene-tissue expression analysis against GTEx v8, and cell-type expression analysis against the DropViz mouse brain atlas and the Siletti/Seeker human brain atlases using the FUMA Cell Type module. Results significant after correction for multiple comparisons in each analysis were evaluated for replication by over-representation analysis (ORA) of the HEAL2 module ($n = 115$ ME/CFS risk genes prioritized by a graph neural network trained on rare coding variants from whole-genome sequencing of 283 cases and 213 controls of European ancestry) against the same gene-set, tissue, and cell-type annotations used in the primary analysis.

2.1. Input Genome-Wide Association Studies

We integrated summary statistics from two publicly available genome-wide association studies (GWAS) of ME/CFS, hereafter referred to as the DecodeME GWAS-1 (DME-1) and Million Veteran Program (MVP) cohorts (Table 2). They were selected among the available summary statistics (Table 1) based on the absence of overlap in controls and the restriction to European ancestry, the latter criterion to ensure the use of a specific reference panel (1000 Genomes Phase 3 European) for linkage disequilibrium (LD) estimation in post-GWAS analyses. Neither DME-1 nor MVP summary statistics include sex chromosomes, so the present study is limited to autosomal variants.

DME-1 comprised 15,579 ME/CFS cases of European ancestry (82% female) and 259,909 sex and ancestry matched controls selected from the UK Biobank (UKB) participants [10]. Summary statistics were downloaded from the Open Science Framework repository and were originally reported on GRCh38. The repository also provides a variant-level quality-control file (`gwas_qced.var.gz`) listing all variants that passed the study-internal filters; this file was used as a pre-filter before applying the standardized thresholds described below. In DecodeME case ascertainment, participants self-reported a clinician-confirmed ME/CFS diagnosis and completed a symptom questionnaire consistent with the Canadian Consensus Criteria [3] and/or the US Institute of Medicine criteria [2].

The MVP cohort comprised 3,891 cases (18.5% female) and 439,202 controls (7.1% female) of European ancestry [27]. Summary statistics were obtained from the NHGRI-EBI GWAS Catalog (accession GCST90479178) and were originally reported on GRCh38. MVP ME/CFS cases were defined using phecode 798.1 (chronic fatigue syndrome) from the PheWAS phecode system [32], which maps to ICD-9 code 780.71(chronic fatigue syndrome), ICD-10-CM code G93.3 (postviral fatigue syndrome), and ICD-10-CM code R53.82 (equivalent to ICD-9 code 780.71). Participants with at least two phecode-mapped ICD code records on distinct dates were classified as cases; those with no such record were classified as controls [27]. Since the specific code G93.32 (myalgic encephalomyelitis/chronic fatigue syndrome) was introduced on October 1, 2022, after the MVP data collection period ended (September 2020) [33], all MVP ME/CFS cases in the present study were necessarily coded under these three pre-2023 codes.

The male prevalence of ME/CFS in the MVP cohort (0.77%) is consistent with the meta-analytic estimate of 0.89% (95% CI 0.60–1.32) across 23 studies on adults [4], while the female prevalence (2.26%) exceeds the corresponding estimate of 1.36% (95% CI 0.91–2.04) [4].

2.2. Quality Control and Harmonization of Summary Statistics

For DME-1, variants were first restricted to those passing the study-internal QC filter (`gwas_qced.var.gz`), which excludes variants with imputation INFO score < 0.9. For both cohorts, variants with minor allele frequency (MAF) < 0.01 were excluded. Summary statistics were then harmonized using the `MungeSumstats` Bioconductor package [31], which performs allele harmonization against dbSNP 155 (GRCh38), strand alignment, duplicate and indel removal, and Z-score computation from β coefficients and standard errors (`compute_z = "BETA"`). Non-biallelic variants were retained (`bi_allelic_filter = FALSE`), with frequency flipping handled as for biallelic sites (`flip_frq_as_biallelic = TRUE`), which means that the computed frequency for flipped non-biallelic sites is overestimated. This affects the estimated β and SE of the meta-analysis but not the p-values, and therefore does not affect any downstream MAGMA analyses (as discussed below). The harmonized GRCh38 files were then lifted over to GRCh37 by a second run of `MungeSumstats` to support downstream FUMA analysis. Correctness of the harmonization was validated by randomly sampling 10,000 variants and verifying that the effect-allele direction and magnitude were preserved (tolerance $|\Delta\beta| < 10^{-6}$, with sign reversal accepted when alleles were complementarily flipped). Because the MVP summary statistics report odds ratios and 95% confidence intervals rather than β coefficients and standard errors, effect-size estimates were derived before harmonization as $\beta = \ln(\mathrm{OR})$, and the standard error was approximated from the 95% confidence interval on the log-odds scale.

2.3. Meta-Analysis

A sample-size-weighted Z-score meta-analysis of DME-1 and MVP ($\sum N_{\mathrm{eff}} = 74{,}219$) was conducted using METAL [34] with the `SCHEME SAMPLESIZE` directive, which combines per-variant Z-scores weighted by effective sample size. The per-variant effective sample size was computed for each cohort as $N_{\mathrm{eff}} = 4\,(N_{\mathrm{cas}}^{-1} + N_{\mathrm{con}}^{-1})^{-1}$ and appended as a dedicated column designated via the `WEIGHTLABEL` directive. Weighted average allele frequencies across cohorts were tracked with the `AVERAGEFREQ ON` directive, and all available variants were included via `REMOVEFILTERS`. Only autosomes were included. METAL recomputes its own per-study Z-scores from p-values and effect direction under the `SCHEME`

Table 2. Summary of GWAS cohorts included in the meta-analysis.

Cohort	N_{cas}	N_{con}	N_{eff}	Variants	% Female	Case definition
DME-1	15,579	259,909	58,792	8,006,900	82.4 (cas), 84.2 (con)	Self-reported clinician-confirmed
MVP	3,891	439,202	15,427	9,147,026	18.5 (cas), 7.1 (con)	EHR-derived (phecode 798.1)

Both cohorts are of European ancestry. Variant counts are after quality control, harmonization, and lift-over to GRCh37 by MungeSumstats [31]. Original assembly: GRCh38 for both cohorts. DME-1: DecodeME main cohort; MVP: Million Veteran Program. $N_{\text{eff}} = 4\,(N_{\text{cas}}^{-1} + N_{\text{con}}^{-1})^{-1}$ denotes the effective sample size. % Female: percentage of female participants among cases (cas) and controls (con); DME-1 values from [10]; MVP values from S.M. Damrauer (personal communication). Case definition: DME-1 cases self-reported a clinician-confirmed ME/CFS diagnosis and met symptom criteria consistent with the Canadian Consensus Criteria and/or US Institute of Medicine criteria. MVP cases were defined by at least two EHR records with ICD codes mapped to phecode 798.1 (chronic fatigue syndrome) [27,32].

`SAMPLESIZE` directive, so the input Z-scores are not directly used in the meta-analysis. Meta-analytic Z-scores are calculated as

$$Z = \frac{Z_{DME\text{-}1}\, w_{DME\text{-}1} + Z_{MVP}\, w_{MVP}}{\sqrt{w_{DME\text{-}1}^2 + w_{MVP}^2}}, \tag{1}$$

where $w_{DME\text{-}1} = \sqrt{N_{\text{eff}}^{(DME\text{-}1)}}$ and $w_{MVP} = \sqrt{N_{\text{eff}}^{(MVP)}}$ are the square roots of the effective sample sizes of DME-1 and MVP, respectively. Meta-analytic p-values are then calculated as $P = 2\,\Phi(-|Z|)$ [34].

For completeness, the meta-analytic summary statistics were post-processed to recover effect-size estimates. Following Equation A.3 of [35], the standard error was approximated as $\text{SE} = (0.5\,N_{\text{eff}}\, f\,(1-f))^{-1/2}$, where f is the weighted average effect-allele frequency and N_{eff} is the meta-analytic effective sample size, and effect sizes were recovered as $\hat{\beta} = Z \times \text{SE}$. These quantities are not used in any downstream analysis; all post-GWAS analyses are based on SNP-level p-values. Note that allele frequencies are affected by overestimation due to the handling of non-biallelic sites during harmonization of the input summary statistics, as described above, which propagates to SE and $\hat{\beta}$ but not to p-values.

The meta-analytic summary statistics were generated from the harmonized GRCh38 summary statistics of both cohorts and then lifted over to GRCh37 with `MungeSumstats` for downstream analysis with FUMA.

2.4. Genomic Locus Definition

SNPs were considered genome-wide significant if $P < 5 \times 10^{-8}$. Among genome-wide significant SNPs, independent significant SNPs were defined as those with pairwise LD $r^2 < 0.6$. Risk loci are built around independent significant SNPs, collecting variants in LD with them at $r^2 \geq 0.6$ and with $P < 0.05$. Adjacent loci separated by less than 250 kb were merged. Within each risk locus, lead SNPs were defined as the most significant SNP among independent significant SNPs that were also mutually independent at $r^2 < 0.1$. Note that this construction allows for more than one independent significant SNP and more than one lead SNP within a single risk locus. LD and MAF were estimated using the 1000 Genomes Project Phase 3 European reference panel. The major histocompatibility complex (MHC) was excluded from locus definition.

2.5. Post-GWAS Analyses with MAGMA

Downstream analyses on the GRCh37 meta-analytic summary statistics were performed using the FUMA platform (v1.8.3) [25] in combination with MAGMA (Multi-marker Analysis of GenoMic Annotation) (v1.10). First, we ran MAGMA gene analysis: variants were mapped to genes using the gene body only (window = 0 kb), and gene-level p-values were computed using the 1000 Genomes Phase 3 European LD reference [24]. For each gene g, a Z-score $z_g = \Phi^{-1}(1 - p_g)$ was then calculated, where Φ^{-1} is the probit function, providing a measure of the association of gene g with ME/CFS. The vector z of gene-level Z-scores was then used as the outcome variable in three linear regression models to detect significant associations between the phenotype and tissues, gene sets, and cell types. The general expression of the regression model is

$$Z \sim \beta_0 + E\beta_1 + C_1\beta_2 + C_2\beta_3 + \ldots + \epsilon, \tag{2}$$

with E representing gene attributes, from binary values reflecting membership in gene sets to expression levels in tissues and cell types, and C_1, C_2, . . . reflecting confounders such as gene length and average gene expression within a dataset [24–26]. Significance is assessed by testing the null hypothesis $\beta_1 = 0$ against the one-sided alternative $\beta_1 > 0$. Bonferroni and Benjamini-Hochberg corrections were applied as described below.

2.5.1. Gene-Set Analysis

In gene-set analysis, the regression in Equation (2) is performed for each of 17,023 gene sets (curated gene sets: 6,494; GO terms: 10,529) from MSigDB (release 2023.1.Hs), with E a binary variable whose value for gene g and set s is 1 if $g \in s$ and 0 otherwise [24]. Gene sets surviving Benjamini-Hochberg (BH) correction over the number of gene sets actually tested by MAGMA ($\alpha = 0.05$) were carried forward to replication. Bonferroni correction results are also reported for comparison. BH-corrected p-values were computed using the `p.adjust` function in R [36] with `method = "BH"`.

2.5.2. Gene-Tissue Expression Analysis

Tissue expression enrichment was assessed using the MAGMA gene-tissue expression module within FUMA SNP2GENE, applied to GTEx v8 expression profiles for 54 tissues. The regression in Equation (2) is performed for each tissue, with E a continuous variable calculated as

$$E_{g,t} = \frac{1}{n_t} \sum_{i=1}^{n_t} \log_2(e_{g,t_i} + 1), \tag{3}$$

where e_{g,t_i} is the expression of gene g in sample t_i of tissue t, and n_t is the total number of samples for that tissue. The average of $E_{g,t}$ across tissues is used as covariate. Tissues surviving Bonferroni correction ($P < 0.05/54 = 9.26 \times 10^{-4}$) were considered significant.

2.5.3. Cell-Type Expression Analysis

Cell-type expression analysis was performed using the FUMA Cell Type module [26] (v1.8.3), taking the MAGMA gene-level association Z-scores from the SNP2GENE job as outcome variable for the regression in Equation (2), performed for each cell type, with the value of E for gene g and cell type c given by

$$E_{g,c} = \frac{1}{n_c} \sum_{i=1}^{n_c} \log_2(e_{g,c_i} + 1), \tag{4}$$

where e_{g,c_i} denotes the raw expression of gene g in cell c_i of cell type c, and n_c is the number of cells in that cell type. The regression includes as covariate the average expression $\bar{A}_g$ of each gene g across all cell types within the same dataset,

$$\bar{A}_g = \frac{1}{N_C} \sum_{c=1}^{N_C} E_{g,c}, \tag{5}$$

where N_C is the total number of cell types in the dataset under investigation [26]. Two independent FUMA jobs were run, one for human brain references and one for mouse brain.

For the human brain, 107 dissection-level datasets were analyzed: 104 from the Siletti et al. (2023) single-nucleus RNA-seq atlas [37] from 4 adult donors (29–60 years), covering cerebral cortex, hippocampus, cerebral nuclei, cerebellum, hypothalamus, midbrain, myelencephalon, pons, thalamus, and spinal cord; and three from the Seeker et al. (2023) white matter panel comprising samples from primary motor cortex (BA4), cerebellum, and cervical spinal cord [38], included to complement the Siletti atlas with white matter cell types not covered by that atlas. Since disease onset in ME/CFS typically occurs within the fourth decade of life [6], we restricted the Seeker datasets to young donors (30–45 years). The total number of cell types across datasets was 2,082.

For the mouse brain, nine region-level datasets from the DropViz atlas [39] were analyzed: Cerebellum (CB), Entopeduncular nucleus and subthalamic nucleus (EP/STN), Frontal cortex (FC), Globus pallidus externus and nucleus basalis (GB/NB), Hippocampus (HP), Posterior cortex (PC), Striatum (STR), Substantia nigra and ventral tegmental area (SN/VTA), and Thalamus (TH). Cell-type annotations were taken from the accompanying DropViz annotation file, downloaded from http://dropviz.org/ [39]. The total number of cell types across datasets was 565.

All datasets were analyzed at level-2 (L2) cell-type resolution. For both mouse and human brain, two parallel analyses were performed: one using Bonferroni correction ($P_{\text{Bon}} < 0.05$, $k = 565$ for mouse; $k = 2{,}082$ for human) and one using Benjamini-Hochberg correction ($P_{\text{BH}} < 0.05$, same k). In both analyses, cell types surviving the significance threshold were carried forward to step 2 of the FUMA cell-type pipeline, in which a forward-selection procedure identifies within-dataset independent signals. Only cell types identified as independent signals in step 2 were considered significant. Subsequently, step 3 conditional analysis was applied to assess independence of signals across datasets. For details on FUMA conditional analyses, see [26].

2.6. Replication of Post-GWAS Analyses

Gene sets, tissues, and cell types surviving correction for multiple testing in the discovery analyses were evaluated for replication using 115 genes previously prioritized by HEAL2, a graph neural network (GNN) that predicts ME/CFS risk from whole-genome sequencing (WGS) data from 283 cases and 213 controls of European ancestry, focusing on loss-of-function and missense variants [20]. Hereafter, this gene set is referred to as the HEAL2 module. The full set of 17,759 genes tested by Zhang et al. (2025) constituted the background for all over-representation analyses (ORA) [20]. Gene symbols were mapped to Ensembl identifiers using the `AnnotationDbi::select` function from the `AnnotationDbi` Bioconductor package [40], with `org.Hs.eg.db` as the annotation database [41]; when a symbol mapped to multiple Ensembl identifiers, all were retained, consistent with the convention used by FUMA to build the MSigDB gene-set annotation file [25]. ORA was performed using the `enricher` function from the `clusterProfiler` Bioconductor package [42], with a one-sided hypergeometric test, `pvalueCutoff = 1`, `qvalueCutoff = 1`, `minGSSize = 0`, and `maxGSSize = Inf` to return results for all tested gene sets regardless of significance. Replication p-values were Bonferroni-corrected for the number of gene sets, tissues, or cell types significant in the discovery analysis, and a gene set, tissue, or cell type was considered replicated at $P_{\text{Bon}} < 0.05$. Foreground gene sets for each analysis type were constructed as described in the following subsections.

2.6.1. Gene-Set Enrichment

ORA was performed via a one-sided hypergeometric test against the same MSigDB GMT collection used in the primary MAGMA analysis (`MSigDB_20231Hs_MAGMA.txt`, available at https://fuma.ctglab.nl/downloadPage), with gene sets and query genes both restricted to the HEAL2 background prior to testing. Bonferroni correction was applied over the number of gene sets with non-empty overlap with the HEAL2 background and therefore testable by ORA.

2.6.2. Tissue Expression Enrichment

The foreground for each tissue comprised the genes significantly upregulated in that tissue relative to all others, as provided in the FUMA tissue expression file `gtex_v8_ts_DEG.txt`. This file was generated by FUMA from GTEx v8 RNA-seq data by two-sided t-tests comparing the $\log_2$ expression of each gene in a given tissue against all other tissues. Genes surviving Bonferroni correction ($\alpha = 0.05$) and absolute $\log_2$ fold-change > 0.58 (equivalent to a fold-change > 1.5) were considered differentially expressed genes (DEGs). The sign of the t-score was used to distinguish between upregulated and downregulated genes [25].

2.6.3. Cell-Type Expression Enrichment

Foreground gene sets were constructed from the FUMA pre-processed single-nucleus RNA-seq expression matrices, downloaded from the FUMA website (https://fuma.ctglab.nl/downloadPage). These matrices are standardized by FUMA from the original atlas data, using log2 transformation with pseudocount, as in Equation (4). The matrices also include the average expression $\bar{A}_g$ of each gene g across all cell types within the same dataset (see Equation (5)) [26]. To identify the specifically expressed genes (SEGs) for each cell type in each dataset, we implemented the top decile expression proportion (TDEP) method, as described in [43]. We converted $E_{g,c}$ to linear scale as $LE_{g,c} = 2^{E_{g,c}} - 1$, and we then computed the expression proportion of gene g in cell type c as

$$PE_{g,c} = \frac{LE_{g,c}}{\sum_{c=1}^{N_C} LE_{g,c}}. \tag{6}$$

where N_C is the number of cell types in the dataset under investigation. Then, we identified as SEGs the genes with $PE_{g,c}$ in the top 10% of all genes in that cell type. We applied as further filter a fold-change threshold of 2, which means that the expression of gene g in cell type c should be at least twice the average expression of gene g across all cell types. We leveraged the log2-transformed expression values $E_{g,c}$ and $\bar{A}_g$ provided by FUMA to apply the fold-change threshold, which is equivalent to applying the following rule: $E_{g,c} - \bar{A}_g > 1$. Gene sets were constructed separately for each cell type in each dataset.

2.7. Software and Reproducibility

All pre-processing and post-processing analyses were conducted in R [36] (v4.4.1). Key packages include `data.table` [44] for large-file I/O, `MungeSumstats` for summary-statistic harmonization and lift-over [31], `clusterProfiler` for over-representation analysis [42], `AnnotationDbi` [40] and `org.Hs.eg.db` [41] for gene-symbol-to-Ensembl identifier mapping. The meta-analysis was performed with METAL (v1.10) [34]. FUMA (v1.8.3) [25,26] and MAGMA (v1.10) [24] were used for locus definition, gene-based, gene-set, tissue expression, and cell-type analyses. The pipeline is controlled by three scripts: `MetaME_main.R`, which downloads and harmonizes the input summary statistics, performs the meta-analysis, and calculates genomic inflation factors; `MetaME_func.R`, which contains all functions called by `MetaME_main.R`; and `MetaME_POST.R`, which reads and processes FUMA output files to perform all post-processing analyses, generates the foreground gene sets for replication, performs all replication analyses, and generates the result tables included in the present manuscript. A configuration file (`MetaME_config.yml`) and two CSV files (`cohorts.csv` and `meta_analyses.csv`) are used to specify the input summary statistics, the desired meta-analysis, and the analysis parameters. This framework can be easily extended to the additional cohorts listed in Table 1 by simply adding rows to `meta_analyses.csv`. It will automatically add METAL correction for overlapping samples, when needed. As soon as new cohorts become available, this framework will provide a basis for future research. Analysis scripts, configuration files, FUMA output, and instructions for reproducing all results are available at https://github.com/paolomaccallini-hub/MetaME_POST under an MIT licence.

3. Results

3.1. Meta-Analysis Identifies Five Genome-Wide Significant Loci

The sample-size-weighted meta-analysis of DME-1 and MVP ($\sum N_{\text{eff}} = 74{,}219$; 8,859,361 variants after post-processing QC and lift-over to GRCh37) identified 158 genome-wide significant variants clustering into five independent genomic risk loci ($P < 5 \times 10^{-8}$; Figure 2a, Table 3). The most significant locus was located on chromosome 20, with three pairwise independent signals (rs4810894, rs8119308, rs3091574), with rs4810894 as the lead ($P = 3.81 \times 10^{-11}$) (Figure 2c). The remaining four loci were located on chromosomes 6 (rs2503773; $P = 3.28 \times 10^{-8}$), 12 (rs12579722; $P = 2.02 \times 10^{-8}$), 15 (rs7165327; $P = 3.11 \times 10^{-8}$), and 19 (rs1982935; $P = 3.85 \times 10^{-8}$), each containing a single independent signal. All but the chromosome 19 locus overlap with loci reported in the original DecodeME preprint, with the chromosome 12 locus detected only in the DME-2 cohort (see Table 1) and below significance in DME-1 [10]. GRCh38 coordinates for lead SNPs are reported in S1 File, sheet S1. Risk loci for DME-1, detected by FUMA SNP2GENE module according to the same criteria, are available in S1 File, sheet S2. No genome-wide significant loci were identified in the MVP cohort.

The meta-analysis showed little evidence of genomic inflation ($\lambda = 1.069$), comparable to that of DME-1 alone ($\lambda = 1.066$) and higher than that of MVP alone ($\lambda = 1.011$). The addition of MVP to DME-1 reduces the association signal for the most significant P values of DME-1, while it improves the signal for less significant P values, as shown by the QQ plot in Figure 2b.

Table 3. Genomic risk loci of the sample-size weighted meta-GWAS.

Lead SNP		Meta (DME-1 + MVP)			DME-1			MVP		
RSID	Variant	Z	FRQ	$-\log_{10} P$	Z	FRQ	$-\log_{10} P$	Z	FRQ	$-\log_{10} P$
rs2503773	6:98,537,145:A:G	−5.53	0.449	**7.48**	−5.29	0.450	6.91	−1.79	0.446	1.14
rs12579722	12:118,563,202:T:C	+5.61	0.061	**7.69**	+4.70	0.061	5.59	+3.14	0.060	2.76
rs7165327	15:55,158,922:A:G	+5.54	0.314	**7.51**	+5.78	0.312	**8.12**	+0.87	0.321	0.41
rs1982935	19:31,857,790:C:T	+5.50	0.181	**7.41**	+4.72	0.180	5.64	+2.84	0.182	2.34
rs4810894†	20:47,532,999:A:G	+6.61	0.633	**10.42**	+6.79	0.634	**10.96**	+1.24	0.628	0.67

FUMA SNP2GENE identified five genomic risk loci for the meta-analysis generated by METAL with the sample-size weighted method, using as input the summary statistics of the main DecodeME cohort and the Million Veteran Program ME/CFS cohort of European ancestry (DME-1 and MVP in Table 1, respectively). Reference build is GRCh37 (GRCh38 coordinates are provided in S1 File, sheet S1). Statistics for the meta-analysis and for the input summary statistics are reported as Z-score (sign reflects effect-allele direction) and $-\log_{10} P$; values exceeding the genome-wide significance threshold $-\log_{10} P > 7.30$ (i.e. $P < 5 \times 10^{-8}$) are in bold. The effect allele is the alternative one. FRQ: effect-allele frequency (weighted average for meta-analysis). †The chromosome 20 locus contains three pairwise independent significant SNPs ($r^2 < 0.6$): rs4810894 (lead), rs8119308, and rs3091574; statistics are shown for the lead SNP only. Risk loci for DME-1 are available in S1 File, sheet S2.

3.2. Gene-Set Enrichment Analysis Reveals Convergent Synaptic Signal

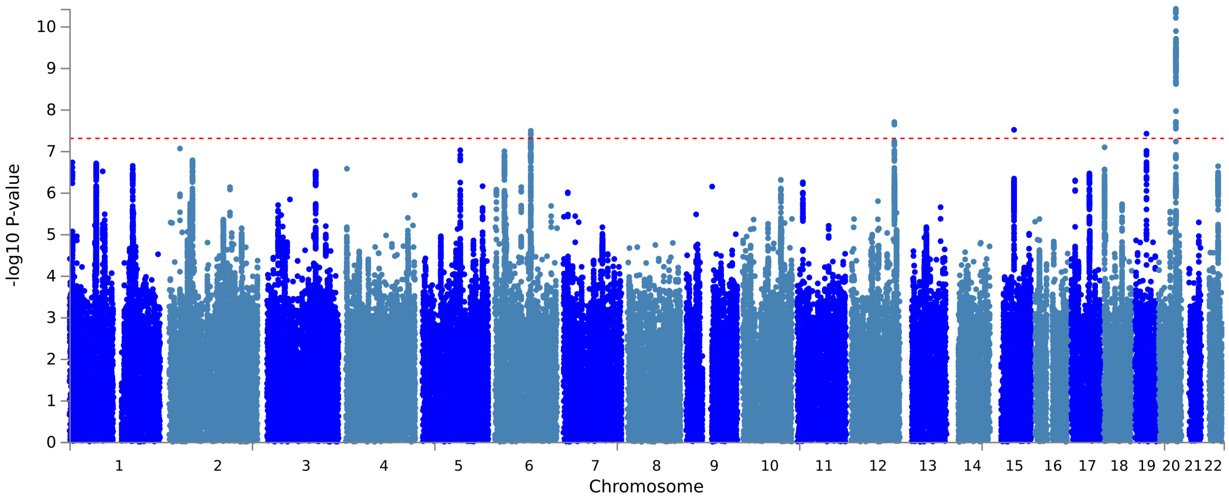
Gene-set analysis using MAGMA tested 17,009 of the 17,023 gene sets in the input file (`MSigDB_20231Hs_MAGMA.txt`). Twelve gene sets were significantly enriched for association signal in the DME-1 + MVP meta-analysis after Benjamini-Hochberg correction ($P_{\mathrm{BH}} < 0.05$, $k = 17{,}009$; Table 4). All 12 span neuronal and synaptic categories from Gene Ontology cellular components, Gene Ontology molecular functions, and Reactome. Three of the 12 also survived the more stringent Bonferroni correction ($P < 0.05/17{,}009 = 2.94 \times 10^{-6}$): postsynaptic specialization, neuron-to-neuron synapse, and somatodendritic compartment, all Gene Ontology cellular component terms implicating the synapse in general and post-synapses in particular. Neither DME-1 nor MVP alone reached significant enrichment after BH correction (S1 File, sheet S3).

Replication was evaluated using the HEAL2 module of ME/CFS risk genes [20]. Only 6,063 of the 17,023 input gene sets had non-empty overlap with the HEAL2 background (17,759 genes) and were testable. Seven of the 12 BH-significant gene sets were replicated at Bonferroni-corrected significance ($P < 0.05/12 = 4.17 \times 10^{-3}$; shaded rows in Table 4), of which six also survived Bonferroni correction in the HEAL2 module ($P < 0.05/6{,}063 = 8.25 \times 10^{-6}$; bold replication p-values in Table 4). All three gene sets surviving Bonferroni correction in discovery were replicated at Bonferroni-corrected significance ($P < 0.05/3 = 0.017$; $P^{\dagger}$ in Table 4).

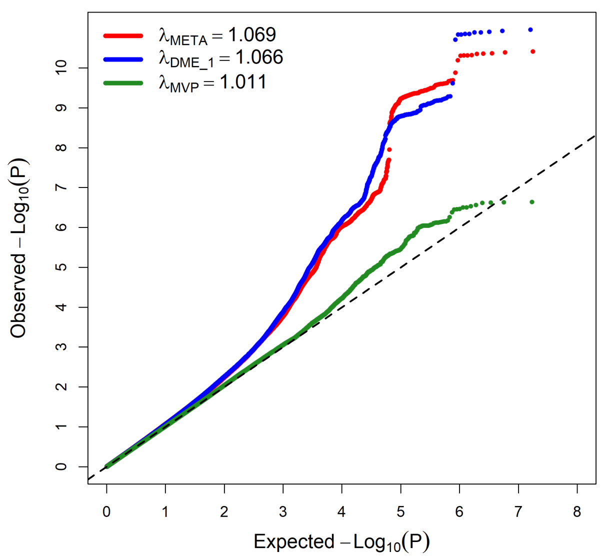
While the three Bonferroni-significant replicated sets point to neuronal soma, dendrites, and post-synapse, BH-significant replicated gene sets extend the signal to the entire synapse, with glutamatergic synapses emerging as the most specific replicated term.

3.3. Tissue Expression Enrichment Implicates Multiple Brain Regions

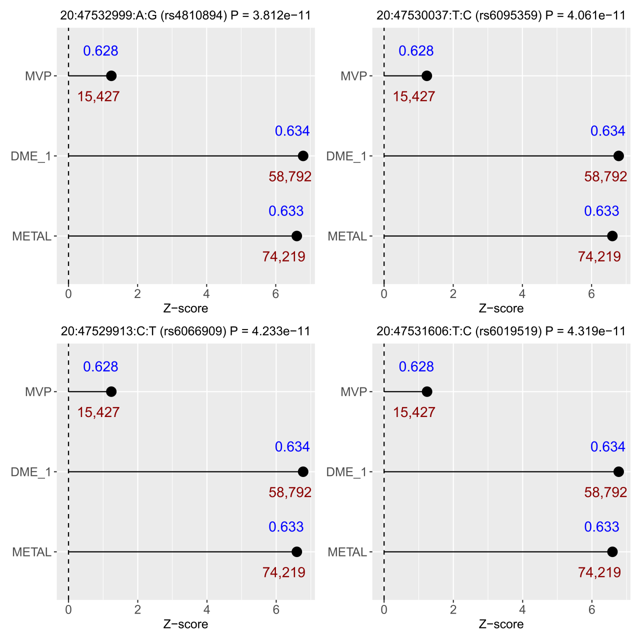
MAGMA gene-tissue expression analysis using GTEx v8 tissue-specific profiles (54 tissues) identified eleven brain regions and the pituitary reaching Bonferroni-corrected significance ($P <$



(**a**)



(**b**)



(**c**)

Figure 2. Genome-wide association results for the DME-1 + MVP meta-analysis. (**a**) Manhattan plot of $-\log_{10} P$ values for 8,859,361 SNPs across the autosomes for the DME-1 + MVP meta-analysis (GRCh37). The horizontal dashed line indicates the genome-wide significance threshold ($P < 5 \times 10^{-8}$). The plot was generated by FUMA [25]. (**b**) Quantile-quantile (QQ) plot of observed versus expected $-\log_{10} P$ values for the DME-1 + MVP meta-analysis (red), DME-1 alone (blue), and MVP alone (green). The dashed line indicates the expected distribution under the null hypothesis of no association. Genomic inflation factors: $\lambda_{META} = 1.069$ for the meta-analysis; $\lambda_{DME-1} = 1.066$ for DME-1; $\lambda_{MVP} = 1.011$ for MVP. (**c**) Z-scores for the four most significant variants of the meta-analysis (all on chromosome 20), shown separately for DME-1 + MVP, DME-1, and MVP. Each panel corresponds to one variant, identified by its GRCh37 coordinates and rs identifier; the meta-analysis p-value is also reported in the panel title. Effect-allele frequencies are annotated in blue above each bar; effective sample sizes are annotated in red below each bar.

Table 4. MAGMA gene-set enrichment results and replication.

		DME-1 + MVP				DME-1			MVP			Replication	
Gene set	N	β	SE	P	P_{BH}	β	SE	P	β	SE	P	FE	P
Postsynaptic specialization[a]	342	0.246	0.047	**9.78×10^{-8}**	0.002	0.171	0.048	1.99×10^{-4}	0.107	0.046	1.01×10^{-2}	7.34	**5.21×10^{-10}**†
Neuron to neuron synapse[a]	360	0.213	0.046	**1.68×10^{-6}**	0.010	0.160	0.047	3.37×10^{-4}	0.072	0.045	5.32×10^{-2}	7.00	**1.05×10^{-9}**†
Somatodendritic compartment[a]	790	0.145	0.031	**1.79×10^{-6}**	0.010	0.121	0.032	7.64×10^{-5}	0.055	0.031	3.56×10^{-2}	3.49	**3.44×10^{-6}**†
Neuronal system[b]	383	0.199	0.046	6.32×10^{-6}	0.027	0.116	0.047	7.11×10^{-3}	0.034	0.045	2.24×10^{-1}	5.68	**1.70×10^{-7}**
Glutamatergic synapse[a]	387	0.184	0.043	9.81×10^{-6}	0.033	0.122	0.045	3.42×10^{-3}	0.064	0.043	6.62×10^{-2}	4.91	**6.15×10^{-6}**
Postsynaptic density membrane[a]	97	0.357	0.087	1.88×10^{-5}	0.046	0.215	0.090	8.38×10^{-3}	0.145	0.086	4.61×10^{-2}	3.15	1.33×10^{-1}
AMPA receptor complex[a]	23	0.711	0.176	2.59×10^{-5}	0.046	0.512	0.178	2.03×10^{-3}	0.473	0.179	4.03×10^{-3}	—	—
Postsynapse[a]	622	0.143	0.035	2.64×10^{-5}	0.046	0.100	0.037	3.10×10^{-3}	0.050	0.035	7.75×10^{-2}	4.79	**1.34×10^{-8}**
Neuron projection[a]	1258	0.100	0.025	3.19×10^{-5}	0.046	0.085	0.026	4.82×10^{-4}	0.060	0.025	7.80×10^{-3}	2.43	1.78×10^{-4}
Inorg. mol. entity transm. transporter[c]	622	0.144	0.036	3.21×10^{-5}	0.046	0.072	0.036	2.34×10^{-2}	0.048	0.035	8.72×10^{-2}	0.49	9.22×10^{-1}
Transmission across synapses[b]	250	0.223	0.056	3.22×10^{-5}	0.046	0.156	0.057	3.42×10^{-3}	−0.002	0.055	5.13×10^{-1}	3.17	2.11×10^{-2}
Synaptic membrane[a]	363	0.185	0.046	3.22×10^{-5}	0.046	0.159	0.048	4.69×10^{-4}	0.033	0.046	2.39×10^{-1}	2.93	1.03×10^{-2}

Regression between gene-level Z-scores for the DME-1 + MVP meta-analysis and gene sets from `MSigDB_20231Hs_MAGMA.txt` identifies 12 regression coefficients (β, corresponding to β_1 of Equation (2)) significantly greater than zero after Benjamini-Hochberg correction ($P_{\text{BH}} < 0.05$). Gene sets span terms from Gene Ontology (GO) cellular components (^{a}), GO molecular functions (^{c}), and Reactome (^{b}). Bonferroni thresholds based on the number of gene sets actually tested by MAGMA in each cohort: $P < 0.05/17{,}009 = 2.94 \times 10^{-6}$ for DME-1 + MVP ($k = 17{,}009$) and MVP ($k = 17{,}009$); $P < 0.05/17{,}008 = 2.94 \times 10^{-6}$ for DME-1 ($k = 17{,}008$). Three gene sets survive Bonferroni correction in DME-1 + MVP ($P < 2.94 \times 10^{-6}$, shown in bold); none survives BH correction in DME-1 or MVP alone (S1 File, sheet S3). Replication was performed as over-representation analysis (ORA) of the HEAL2 module ($n = 115$ ME/CFS risk genes) [20] against the same MSigDB gene set collection, restricting foreground and background to the 17,759 genes used in the original HEAL2 study [20], implemented via `enricher` in `clusterProfiler` [42]. Of the 17,023 input gene sets, $k = 6{,}063$ had non-empty overlap with the HEAL2 background and were testable; Bonferroni correction for genome-wide HEAL2 replication was applied over these $k = 6{,}063$ tests ($P < 0.05/6{,}063 = 8.25 \times 10^{-6}$). The AMPA receptor complex (—) had no HEAL2 genes in the background and could not be tested. Shaded rows indicate positive replication ($P < 0.05/12 = 4.17 \times 10^{-3}$) of gene sets surviving BH correction in discovery. $P^{\dagger}$ indicates gene sets surviving Bonferroni correction in discovery ($P < 2.94 \times 10^{-6}$) that were replicated at Bonferroni-corrected significance in the HEAL2 module ($P < 0.05/3 = 0.017$). N: number of genes in the gene set. SE: standard error of β. FE: fold enrichment. P: raw p-value.

$0.05/54 = 9.26 \times 10^{-4}$; Table 5; Figure 3). The strongest signals were observed in cerebellar hemisphere ($P = 2.87 \times 10^{-7}$; $P_{\mathrm{Bon}} = 1.55 \times 10^{-5}$; $\beta = 0.031$) and cerebellum ($P = 3.50 \times 10^{-7}$; $P_{\mathrm{Bon}} = 1.89 \times 10^{-5}$; $\beta = 0.031$), followed by frontal cortex BA9 ($P = 3.81 \times 10^{-7}$), cortex ($P = 4.64 \times 10^{-7}$), nucleus accumbens basal ganglia ($P = 2.58 \times 10^{-6}$), anterior cingulate cortex BA24 ($P = 5.34 \times 10^{-6}$), caudate basal ganglia ($P = 1.51 \times 10^{-5}$), hypothalamus ($P = 1.78 \times 10^{-5}$), hippocampus ($P = 6.09 \times 10^{-5}$), amygdala ($P = 6.53 \times 10^{-5}$), putamen basal ganglia ($P = 1.03 \times 10^{-4}$), and pituitary ($P = 2.37 \times 10^{-4}$). No peripheral tissue approached significance. All significant tissues were significant in DME-1 alone, with comparable effect sizes (β) and generally smaller P-values, while MVP showed no significant tissue enrichment (Table 5 and S1 File, sheet S4).

Replication was performed by over-representation analysis (ORA) of the HEAL2 module ($n = 115$ genes) against genes upregulated in each significant tissue (GTEx v8 `gtex_v8_ts_DEG.txt`), using as background the 17,759 genes investigated in the original HEAL2 study, implemented via the `enricher` function in `clusterProfiler` [42]. Three tissues survived Bonferroni correction for replication ($P < 0.05/12 = 4.17 \times 10^{-3}$): frontal cortex BA9 ($P = 1.57 \times 10^{-3}$), nucleus accumbens basal ganglia ($P = 1.77 \times 10^{-3}$), and hippocampus ($P = 2.05 \times 10^{-3}$). In the original HEAL2 study, enrichment against 50 human tissues using a two-sided t-test on expression levels reported 24 Bonferroni-significant signals, with cerebral cortex, basal ganglia, hippocampus, hypothalamus, cerebellum, amygdala, and pituitary among the top-ten results [20]; a profile closely resembling our discovery results.

3.4. Human Brain Cell-Type Analysis Implicates Eccentric Medium Spiny Neurons and Glutamatergic Neurons in White Matter

Cell-type expression analysis using the FUMA Cell Type module applied to the Siletti et al. (2023) human brain single-nucleus RNA-seq atlas [37] and the Seeker et al. (2023) white-matter panel [38] (107 datasets; level-2 resolution; 2,082 cell-type tests in total) identified one cell type surviving Bonferroni correction ($P < 0.05/2{,}082 = 2.40 \times 10^{-5}$): the eccentric medium spiny neuron (eMSN) in the claustrum (S-47; $P = 7.93 \times 10^{-6}$; $P_{\mathrm{Bon}} = 0.017$; $\beta = 0.045$, SE $= 0.011$; Table 6). No cell type in the Seeker white-matter datasets reached Bonferroni significance.

To characterise the broader association signal, the analysis was extended using Benjamini-Hochberg (BH) correction ($k = 2{,}082$), which identified 19 independent signals passing $P_{\mathrm{BH}} < 0.05$ and FUMA step 2 conditional analysis (Table 6). Fourteen corresponded to eMSNs distributed across multiple cerebral nuclei regions in the Siletti atlas, including the claustrum, globus pallidus externus, septum, substantia innominata, central nucleus, putamen, basolateral and corticomedial nuclear groups of the amygdala, and the insular cortex, as well as two thalamic subregions (VA, VLN) and two

Table 5. MAGMA gene-tissue expression analysis and replication.

	DME-1 + MVP				DME-1			MVP			Replication	
Tissue	β	SE	P	P_{Bon}	β	SE	P	β	SE	P	FE	P
Cerebellar hemisphere	0.031	0.006	2.87×10^{-7}	1.55×10^{-5}	0.026	0.006	$\mathbf{1.97 \times 10^{-5}}$	−0.006	0.006	8.38×10^{-1}	1.36	3.74×10^{-2}
Cerebellum	0.031	0.006	3.50×10^{-7}	1.89×10^{-5}	0.027	0.007	$\mathbf{2.37 \times 10^{-5}}$	−0.006	0.006	8.21×10^{-1}	1.43	2.18×10^{-2}
Frontal cortex BA9	0.034	0.007	3.81×10^{-7}	2.06×10^{-5}	0.039	0.007	$\mathbf{3.37 \times 10^{-8}}$	0.002	0.007	3.60×10^{-1}	1.90	$1.57 \times 10^{-3\dagger}$
Cortex	0.035	0.007	4.64×10^{-7}	2.51×10^{-5}	0.039	0.007	$\mathbf{5.71 \times 10^{-8}}$	0.005	0.007	2.56×10^{-1}	1.64	2.45×10^{-2}
Nucleus accumbens (BG)	0.035	0.008	2.58×10^{-6}	1.39×10^{-4}	0.039	0.008	$\mathbf{4.40 \times 10^{-7}}$	0.010	0.008	9.38×10^{-2}	2.08	$1.77 \times 10^{-3\dagger}$
Anterior cingulate cortex BA24	0.032	0.007	5.34×10^{-6}	2.88×10^{-4}	0.040	0.008	$\mathbf{6.91 \times 10^{-8}}$	0.001	0.007	4.46×10^{-1}	1.58	5.55×10^{-2}
Caudate basal ganglia	0.033	0.008	1.51×10^{-5}	8.14×10^{-4}	0.038	0.008	$\mathbf{1.29 \times 10^{-6}}$	0.008	0.008	1.46×10^{-1}	1.85	1.32×10^{-2}
Hypothalamus	0.034	0.008	1.78×10^{-5}	9.63×10^{-4}	0.038	0.008	$\mathbf{2.67 \times 10^{-6}}$	0.001	0.008	4.58×10^{-1}	1.62	3.93×10^{-2}
Hippocampus	0.031	0.008	6.09×10^{-5}	3.29×10^{-3}	0.037	0.008	$\mathbf{3.75 \times 10^{-6}}$	0.002	0.008	3.93×10^{-1}	2.18	$2.05 \times 10^{-3\dagger}$
Amygdala	0.030	0.008	6.53×10^{-5}	3.53×10^{-3}	0.037	0.008	$\mathbf{2.25 \times 10^{-6}}$	0.000	0.008	4.99×10^{-1}	1.61	7.66×10^{-2}
Putamen basal ganglia	0.030	0.008	1.03×10^{-4}	5.56×10^{-3}	0.036	0.008	$\mathbf{7.41 \times 10^{-6}}$	0.006	0.008	2.31×10^{-1}	1.39	1.95×10^{-1}
Pituitary	0.032	0.009	2.37×10^{-4}	1.28×10^{-2}	0.030	0.009	$\mathbf{5.76 \times 10^{-4}}$	−0.003	0.009	6.11×10^{-1}	1.23	1.86×10^{-1}

Tissues surviving family-wise error correction (FWER), using Bonferroni correction ($P < 0.05/54 = 9.26 \times 10^{-4}$), in the DME-1 + MVP meta-analysis are ordered by significance. Per-cohort results are also shown for comparison, along with replication results. Regression coefficients and corresponding standard errors of MAGMA tissue-type analysis are indicated by β and SE, respectively. See Equation (2) for reference. All tissues reaching Bonferroni-corrected significance in the meta-analysis also reached Bonferroni-corrected significance in DME-1 alone (bold P values), while none reached nominal significance in MVP. The complete results across all 54 regressions for DME-1 + MVP, DME-1, and MVP are collected in S1 File, sheet S4. Replication was performed by over-representation analysis of the HEAL2 module ($n = 115$ genes) against genes upregulated in each significant tissue (GTEx v8 `gtex_v8_ts_DEG.txt`), using the 17,759 background genes employed in the original HEAL2 study, implemented via the `enricher` function in `clusterProfiler` [42]. $P^{\dagger}$ indicates gene sets surviving Bonferroni correction in discovery that were replicated at Bonferroni-corrected significance in the HEAL2 module ($k = 12$; threshold $P < 0.05/12 = 4.17 \times 10^{-3}$). FE: fold enrichment.

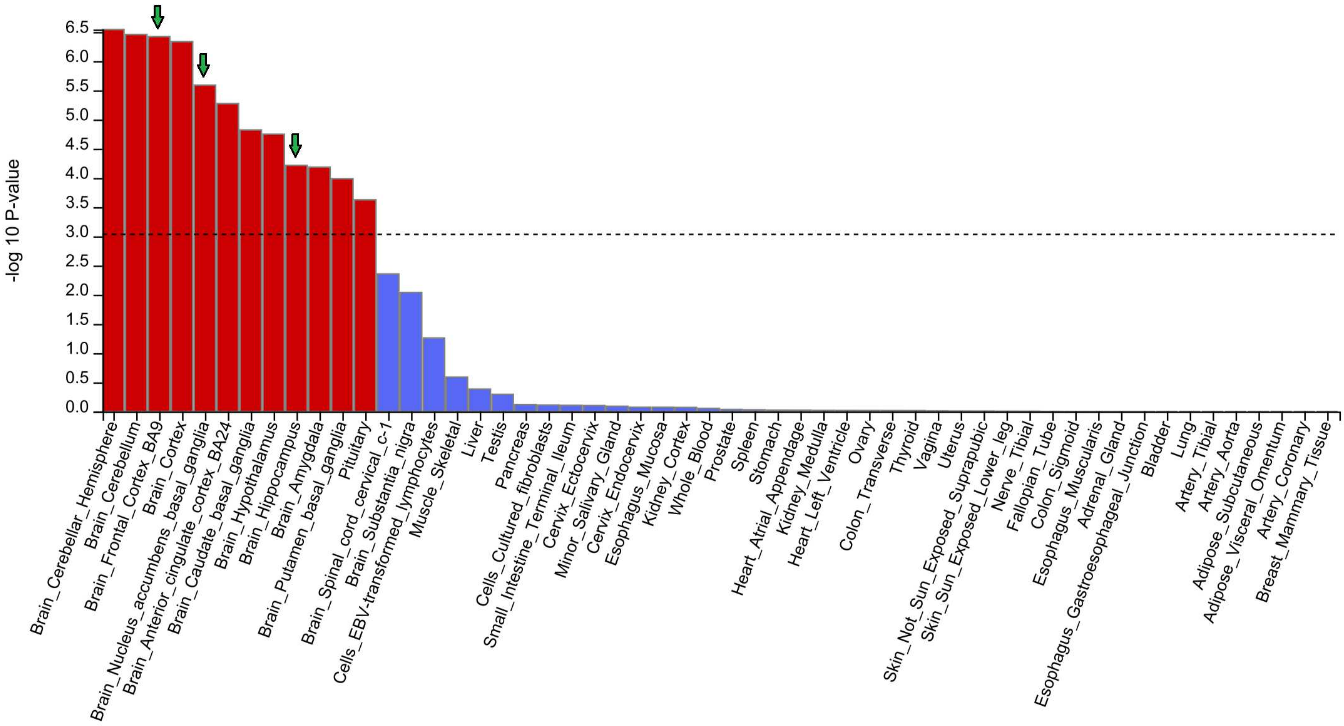


Figure 3. MAGMA gene property analysis for tissue specificity. Bar plot of $-\log_{10} P$ values for the association between tissue-specific gene expression profiles and DME-1 + MVP gene-level association statistics across 54 GTEx v8 tissues, computed using MAGMA v1.10 gene-property analysis within the FUMA SNP2GENE pipeline [25]. The dashed line indicates the Bonferroni significance threshold ($P < 0.05/54 = 9.26 \times 10^{-4}$, corresponding to $-\log_{10} P > 3.03$). Of the twelve tissues reaching Bonferroni-corrected significance, eleven are brain regions; the pituitary also reached significance. No peripheral tissue reached significance. All twelve significant tissues were also individually significant in DME-1 alone (Bonferroni correction, $k = 54$), while none was associated with MVP alone. Downward arrows indicate tissues that survived Bonferroni-corrected replication against the HEAL2 module ($P < 0.05/12 = 4.17 \times 10^{-3}$; see Table 5).

hippocampal subfields (HiH.CA1, HiT.CA4). One additional signal corresponded to a medium spiny neuron (MSN, distinct from eMSN) in the hippocampus head CA1 field (S-61). The remaining four signals comprised three glutamatergic neurons in the Seeker white-matter panel (SK-332 BA4, SK-334 CB, SK-336 CSC) and one thalamic excitatory neuron (S-106 LP.VPL). The extensive collinearity among eMSN entries (Figure 4) indicates a single distributed genetic signal rather than independent cell-type associations.

Replication was evaluated using the HEAL2 module of ME/CFS risk genes [20], implemented via the `enricher` function in `clusterProfiler` [42]. Of the 19 BH-significant signals, two were replicated at Bonferroni-corrected significance ($P < 0.05/19 = 2.63 \times 10^{-3}$; shaded rows in Table 6): the eMSN in the basolateral nuclear group lateral nucleus (S-57 BLN.La; $P = 2.48 \times 10^{-3}$; FE $= 2.31$) and the glutamatergic neuron in cerebellar white matter (SK-334 CB; $P = 1.85 \times 10^{-3}$; FE $= 3.47$). The glutamatergic cerebellar signal is independent from the eMSN signal, representing a distinct biological finding, according to FUMA step 3 conditional analysis (Figure 4). Genes from the HEAL2 modules that overlap with cell SEGs are collected in the supplementary file, for each one of the cell types tested (S1 File, sheet S5). Neither DME-1 nor MVP alone reached BH significance for any cell type in the human atlas (S1 File, sheet S5).

3.5. Mouse Cell-Type Expression Analysis Was Not Replicated

Cell-type expression analysis using the FUMA Cell Type module with the DropViz mouse brain atlas (nine regions; level-2 resolution) identified six neuronal cell types surviving Bonferroni correction

Table 6. Independent cell types from step 2 of the FUMA cell-type pipeline (Siletti/Seeker human brain atlases; level-2 resolution; BH correction), with per-cohort statistics and HEAL2 module replication.

				DME-1 + MVP				DME-1			MVP			Replication	
Atlas	Gross region	Dissection	Cell type	β	SE	P	P_{BH}	β	SE	P	β	SE	P	FE	P
S-47	Cerebral nuclei	Cla	eMSN	0.045	0.011	$\mathbf{7.93 \times 10^{-6}}$	0.017	0.027	0.011	7.09×10^{-3}	0.023	0.010	1.48×10^{-2}	1.44	1.35×10^{-1}
S-44	Cerebral nuclei	GP.Gpe	eMSN	0.033	0.008	3.20×10^{-5}	0.030	0.021	0.009	6.85×10^{-3}	0.015	0.008	3.53×10^{-2}	1.16	3.49×10^{-1}
SK-332	White matter	BA4	Glut.	0.034	0.009	6.23×10^{-5}	0.030	0.035	0.009	4.62×10^{-5}	0.012	0.009	7.51×10^{-2}	1.64	9.28×10^{-2}
S-46	Basal forebrain	SEP	eMSN	0.035	0.009	1.09×10^{-4}	0.030	0.018	0.010	3.41×10^{-2}	0.018	0.009	2.23×10^{-2}	1.33	2.72×10^{-1}
S-49	Basal forebrain	SI	eMSN	0.038	0.010	1.13×10^{-4}	0.030	0.017	0.011	5.99×10^{-2}	0.024	0.010	9.80×10^{-3}	2.05	3.97×10^{-3}
S-45	Amygdala	CEN	eMSN	0.040	0.011	1.24×10^{-4}	0.030	0.025	0.011	1.46×10^{-2}	0.025	0.011	9.33×10^{-3}	1.79	2.20×10^{-2}
S-52	Basal nuclei	Pu	eMSN	0.033	0.009	1.28×10^{-4}	0.030	0.022	0.009	8.56×10^{-3}	0.010	0.009	1.18×10^{-1}	1.75	1.75×10^{-2}
S-57	Amygdala	BLN.La	eMSN	0.042	0.012	1.36×10^{-4}	0.030	0.028	0.012	1.03×10^{-2}	0.026	0.011	1.07×10^{-2}	2.31	2.48×10^{-3}
S-54	Basal nuclei	CaB	eMSN	0.028	0.008	1.47×10^{-4}	0.030	0.018	0.008	1.12×10^{-2}	0.010	0.008	9.25×10^{-2}	1.89	1.17×10^{-2}
S-48	Amygdala	CMN	eMSN	0.046	0.013	1.60×10^{-4}	0.030	0.029	0.013	1.46×10^{-2}	0.029	0.013	9.70×10^{-3}	1.30	2.74×10^{-1}
S-110	Thalamus	VA	eMSN	0.032	0.009	1.80×10^{-4}	0.030	0.017	0.009	3.45×10^{-2}	0.015	0.009	4.76×10^{-2}	1.71	2.68×10^{-2}
SK-334	White matter	CB	Glut.	0.043	0.012	1.87×10^{-4}	0.030	0.043	0.013	3.44×10^{-4}	0.020	0.012	4.95×10^{-2}	3.47	1.85×10^{-3}
S-106	Thalamus	LP.VPL	Thal.exc.	0.025	0.007	2.19×10^{-4}	0.033	0.018	0.007	5.71×10^{-3}	0.004	0.007	2.60×10^{-1}	0.99	5.66×10^{-1}
S-61	Hippocampus	HiH.CA1	MSN	0.033	0.010	3.34×10^{-4}	0.044	0.024	0.010	7.51×10^{-3}	0.014	0.009	6.90×10^{-2}	1.06	4.64×10^{-1}
S-15	Cerebral cortex	Ig	eMSN	0.014	0.004	3.45×10^{-4}	0.044	0.005	0.004	9.26×10^{-2}	0.001	0.004	3.76×10^{-1}	0.89	7.45×10^{-1}
S-59	Hippocampus	HiT.CA4	eMSN	0.026	0.008	3.73×10^{-4}	0.044	0.019	0.008	8.48×10^{-3}	0.007	0.008	1.88×10^{-1}	1.35	1.71×10^{-1}
S-50	Amygdala	BLN.BL	eMSN	0.038	0.011	3.84×10^{-4}	0.044	0.022	0.012	2.72×10^{-2}	0.025	0.011	1.08×10^{-2}	1.94	8.82×10^{-3}
SK-336	White matter	CSC	Glut.	0.038	0.012	4.24×10^{-4}	0.046	0.042	0.012	2.32×10^{-4}	0.016	0.011	8.09×10^{-2}	2.41	2.49×10^{-2}
S-100	Thalamus	VLN	eMSN	0.028	0.008	4.38×10^{-4}	0.046	0.020	0.009	1.06×10^{-2}	0.008	0.008	1.67×10^{-1}	2.00	2.09×10^{-2}

Nineteen cell types passed Benjamini-Hochberg correction ($P_{BH} < 0.05$, $k = 2{,}082$) and FUMA step 2 conditional analysis in the DME-1 + MVP meta-analysis. One also passed the more stringent Bonferroni correction (bold P value): the claustrum eccentric medium spiny neuron (S-47 Cla eMSN). Bonferroni thresholds based on the number of cell types actually tested by MAGMA in each cohort: $P < 0.05/2{,}082 = 2.40 \times 10^{-5}$ for DME-1 + MVP, DME-1, and MVP ($k = 2{,}082$ for all three cohorts). Neither DME-1 nor MVP alone reached BH significance for any cell type. The complete results for all cohorts and replication are provided in S1 File, sheet S5. Replication was performed as over-representation analysis of the HEAL2 module ($n = 115$ genes) [20] against cell-type foreground gene sets (background: 17,759 genes from the HEAL2 study), implemented via the `enricher` function in `clusterProfiler` [42]. Of the 2,082 cell types tested in discovery, $k = 2{,}052$ had non-empty overlap with the HEAL2 background and were testable in replication ($P < 0.05/2{,}052 = 2.44 \times 10^{-5}$ for genome-wide HEAL2 replication). Shaded rows indicate positive replication ($P < 0.05/19 = 2.63 \times 10^{-3}$) of cell types surviving BH correction in discovery. S-n: Siletti et al. (2023) [37], dataset number n; SK-n: Seeker et al. (2023) [38] (young donors, 30–45 years), dataset number n. Dissection abbreviations (Siletti atlas): Cla: claustrum; GP.Gpe: external segment of globus pallidus; SEP: septal nuclei; SI: substantia innominata and nearby nuclei; CEN: central nuclear group of the amygdala; Pu: putamen; BLN.La: lateral nucleus of the amygdaloid complex (basolateral nuclear group); CaB: body of the caudate nucleus; CMN: corticomedial nuclear group of the amygdala; VA: ventral anterior nucleus of thalamus (lateral nuclear complex); LP.VPL: lateral posterior + ventral posterior lateral nucleus of thalamus (lateral nuclear complex); VLN: ventral group of lateral nuclear complex of thalamus; HiH.CA1: head of hippocampus, uncal CA1; Ig: granular insular cortex; HiT.CA4: tail of hippocampus, caudal CA4-DGC; BLN.BL: basal nucleus of the amygdaloid complex (basolateral nuclear group). Seeker atlas dissections: BA4: primary motor cortex; CB: cerebellum; CSC: cervical spinal cord. Cell-type abbreviations: eMSN: eccentric medium spiny neuron; MSN: medium spiny neuron; Glut.: glutamatergic neuron; Thal.exc.: thalamic excitatory neuron. Statistical abbreviations: β: MAGMA regression coefficient; SE: standard error; P: raw p-value; P_{BH}: Benjamini-Hochberg corrected p-value ($k = 2{,}082$); FE: fold enrichment. Anatomical nomenclature follows Siletti et al. (2023) [37].

($P < 0.05/565 = 8.85 \times 10^{-5}$) and FUMA step 2 conditional analysis in the DME-1 + MVP meta-analysis (Table A1). The strongest signal arose in the striatum (STR), where a GABAergic *Drd1*$^{+}$ direct-pathway spiny projection neuron (dSPN) reached $P = 3.96 \times 10^{-8}$ ($P_{Bon} = 2.24 \times 10^{-5}$; $\beta = 0.038$). A second independent signal was identified in the globus pallidus (GP), where a GABAergic tyrosine hydroxylase-positive (*Th*$^{+}$) spiny projection neuron expressing *Adora2a* reached $P = 1.76 \times 10^{-7}$ ($P_{Bon} = 9.95 \times 10^{-5}$; $\beta = 0.037$). Four further independent signals were identified in the substantia nigra (SN; a glutamatergic *Slc17a6*$^{+}$ neuron), cerebellum (CB; a glutamatergic *Gabra6*$^{+}$ granule cell), posterior cortex (PC; a glutamatergic *Syt6*$^{+}$ layer 6a neuron), and hippocampus (HC; a glutamatergic dentate principal cell expressing *C1ql2*). Step 3 conditional analysis identified three collinear pairs among the six Bonferroni-significant signals: the Th^{+} SPN (GP) and the glutamatergic neuron (SN); the granule cell (CB) and the layer 6a neuron (PC); and the layer 6a neuron (PC) and the dentate principal cell (HC) (Figure A1).

Under the less stringent Benjamini-Hochberg correction, 21 within-dataset independent cell types reached statistical significance, spanning all nine DropViz regions and including both GABAergic and glutamatergic subtypes (Table A1). DME-1 alone reached Bonferroni significance for five of the BH-significant cell types, including the *Th*$^{+}$ SPN in GP and the dentate principal cell (*C1ql2*$^{+}$) in HC; MVP alone showed no significant signals (S1 File, sheet S6). None of the 21 BH-significant cell types was replicated.

The DropViz atlas uses the term spiny projection neuron (SPN) for the principal GABAergic neurons of the striatum, equivalent to the medium spiny neuron (MSN) terminology used in the human atlas. Eccentric SPNs (eSPNs) are the mouse homologues of human eMSNs: the *eccentric* supercluster of SPNs was first identified in mice by Saunders et al. [39] and subsequently recognized in humans by Siletti et al. [37]. The dSPN and *Th*$^{+}$ SPN signals identified here are striatal subtypes distinct from eSPNs and therefore do not correspond to the eMSN signal identified in the human brain analysis (Table 6). We note, however, that the *Th*$^{+}$ SPN (GP dataset, subcluster 3-9) shares the *Adora2a-Th* type marker with the eccentric SPN subtype 13-5 of the striatum dataset [39], which the DropViz curators explicitly label as eccentric in the annotation; the relationship between these two populations warrants further investigation. An eSPN subtype in the striatum dataset (subcluster 13-2, dSPN-like markers; $P = 8.72 \times 10^{-6}$; $P_{\mathrm{Bon}} = 0.005$) also reached Bonferroni-corrected significance in step 1 (S1 File, sheet S6) but was removed by the FUMA step 2 conditional analysis, indicating collinearity with the dSPN signal, consistent with the transcriptional overlap between eSPNs and canonical SPNs documented by Saunders et al. [39].

4. Discussion

In this analysis of a meta-GWAS of 19,470 ME/CFS cases and 699,111 controls, we revealed heritability enriched in brain tissues, in neuronal cell types, and in gene sets pointing to synapses and post-synapses. We provided independent replication of these results on 115 ME/CFS risk genes previously associated with the disease by WGS prioritization of rare variants. The meta-analysis was necessary to detect the signal in gene-set and cell-type analysis for the human brain: neither MVP nor DME-1 alone reached Benjamini-Hochberg (BH) significance for either of the two analyses. Replication of the post-GWAS findings through over-representation analysis of the HEAL2 ME/CFS risk gene module extends to ME/CFS the previously documented interplay between genes harbouring common variants and those harbouring rare variants in complex traits [17,18].

The replicated brain tissues implicated the frontal cortex (BA9), basal ganglia, and hippocampus while gene-set analysis pointed to post-synapses, with glutamatergic synapses as the most specific signal. These results prompted the focus on cell-type analysis in human brain, which revealed two independent cell types: eccentric medium spiny neurons (eMSNs) in the lateral nucleus of the amygdala (BLN.La) and a population of glutamatergic neurons in cerebellar white matter.

The function of eMSNs is not well understood and they have only recently been discovered in mouse and human brain [37,39]. They are a transcriptionally distinct class of medium spiny neurons

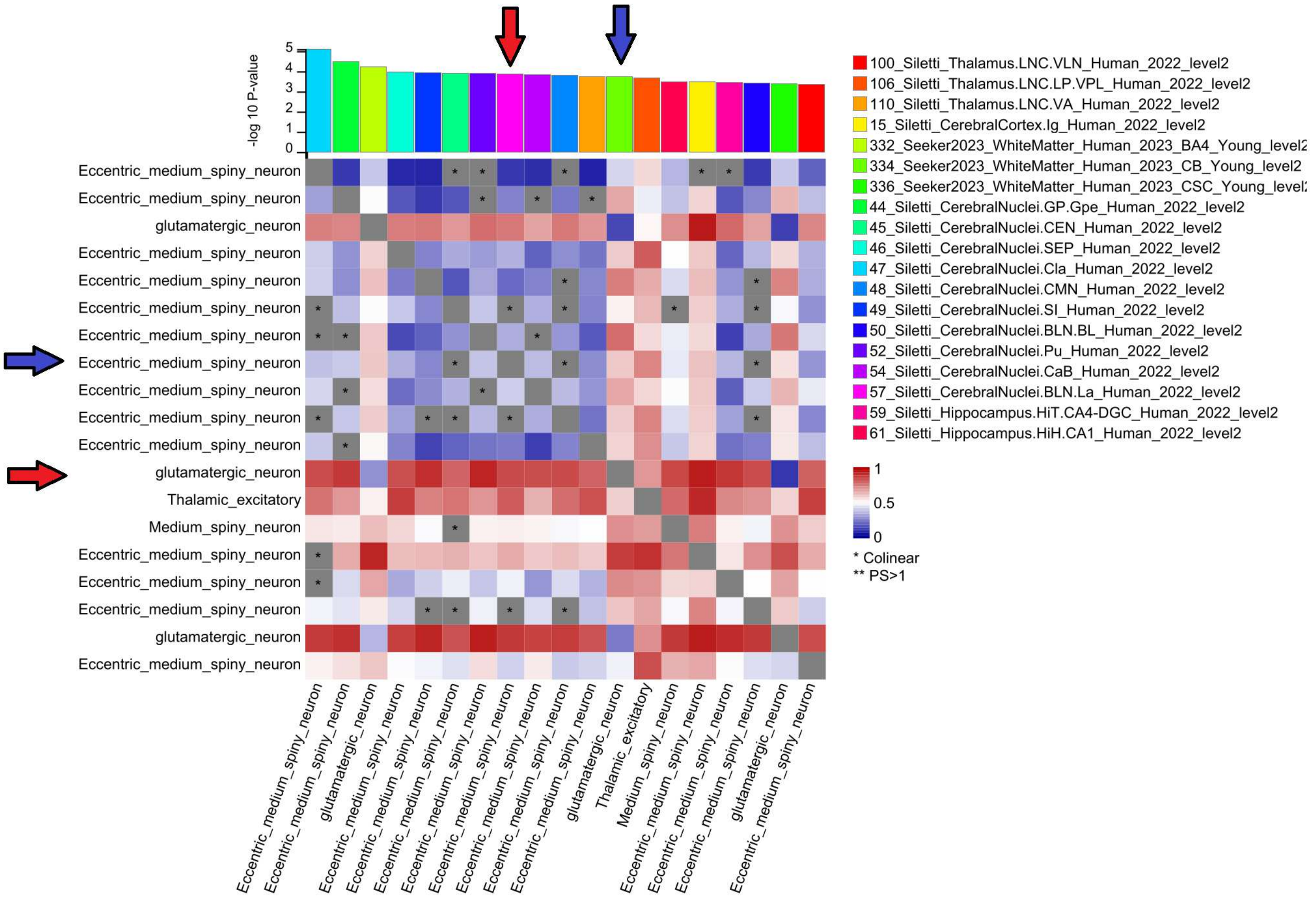


Figure 4. Step 3 conditional analysis from the FUMA cell-type pipeline (Siletti/Seeker human brain atlases; level-2 resolution; BH correction; DME-1 + MVP meta-analysis). Cell types that survive BH correction in discovery and FUMA step 2 conditional analysis (Table 6) were subjected to cross-dataset conditional analysis for the detection of collinearity (FUMA step 3). The bar plot (top) shows the marginal $-\log_{10} P$ for each of the 19 independent cell types on the x-axis. The heatmap shows pairwise cross-dataset proportional significance (PS): the value in row i, column j is the $-\log_{10} P$ of cell type j after conditioning on cell type i. Values close to 1 (red) indicate that the signal of cell type j is retained after conditioning on i (independent signals); values close to 0 (blue) indicate attenuation. Starred cells (*) denote collinear pairs [26]. Cells are displayed in decreasing order of significance, as in Table 6. We see extensive collinearity among eccentric medium spiny neuron (eMSN) entries across cerebral nuclei. Two signals survived Bonferroni-corrected replication in the HEAL2 module: the eMSN in the basolateral amygdala lateral nucleus (BLN.La) and the glutamatergic neuron in cerebellar white matter (WM.CB). The WM.CB signal (vertical blue arrow) is retained after conditioning on the eMSN signal (horizontal blue arrow), and vice versa (red arrows). Since $PS > 0.5$ in both comparisons, the two signals are considered independent according to the FUMA step 3 criteria.

(MSNs), which are the principal cell type of the striatum. Like MSNs in general, eMSNs are GABAergic neurons with dopaminergic and glutamatergic input [45], but unlike canonical MSNs, eMSNs are more broadly and evenly distributed across the cerebral nuclei [37,39]. Recent cell-type analyses in the context of post-GWAS interpretation have implicated eMSNs in educational attainment and intelligence, and in neuropsychiatric disorders [43]. Our results may provide a genetic basis for the consistent finding of additional brain area recruitment during cognitive tasks [9] and for the almost ubiquitous observation of cognitive dysfunction in this patient population [10], characterized by deficits in attention, memory, and processing speed [8].

The other cell type implicated by the present analysis is a population of interstitial white matter neurons (IWMNs) in the cerebellum. Abnormalities in IWMNs have been documented in several brain diseases, and a possible role in brain wiring, microcirculation, and myelination has been suggested [46]. In ME/CFS, white matter microstructural abnormalities have been reported across multiple fibre tracts, including cerebellar peduncles, and correlate with mental health and disability outcomes [47]. Whether IWMNs contribute to these abnormalities is unknown, since the investigations on the role of IWMNs in human diseases has required specific postmortem studies [46], never attempted in ME/CFS, to our knowledge.

The replicated gene-set signals suggest a role for postsynaptic compartments and glutamatergic synapses in ME/CFS pathogenesis. Combined with the cell-type findings in the human brain, these associations may reflect in particular the glutamatergic input onto eMSNs and the glutamatergic identity of IWMNs in the cerebellum. Convergent neurochemical evidence comes from magnetic resonance spectroscopy (MRS) studies documenting elevated glutamate levels in the posterior cingulate cortex in both ME/CFS and Long COVID [48], and from a report of widespread AMPA receptor upregulation in Long COVID patients with objective cognitive impairment [49], a condition showing substantial overlap with ME/CFS [16]. Glutamatergic dysregulation may also explain the unusual prevalence of alcohol intolerance in ME/CFS [7], since alcohol inhibits NMDA receptor-mediated transmission [50]. At the monogenic level, the best-documented case involves a structural variant spanning *AKR1C1* and *AKR1C2* that disrupts expression of both genes and increases synthesis of inhibitory GABAergic neurosteroids [21], pointing to the GABAergic side of the same excitatory–inhibitory balance. Together, GWAS, neuroimaging, and monogenic evidence converge on an imbalance between excitatory and inhibitory neurotransmission as a candidate mechanism in ME/CFS pathogenesis.

The cell-type analysis of the DropViz mouse brain atlas identified significant signals in the discovery phase with only partial overlap with the human brain signals: in both analyses, the significant cell types were exclusively neuronal, and MSNs were implicated in both species, but the specific cell types and their regional distribution differed. This discrepancy may reflect the documented transcriptional divergence between homologous mouse and human brain cell types [51]. This transcriptional divergence may also explain why the cell-type enrichment results from the DropViz mouse brain atlas were not replicated by the HEAL2 module.

Several limitations should be noted. The MVP cases were ascertained using PheCode 798.1 from electronic health records, which may capture a more heterogeneous phenotype than the clinician-confirmed CCC/IOM diagnosis used in DecodeME. The HEAL2 gene module used for replication is derived from a preprint that has not yet undergone peer review. An overlap between DecodeME cases and CureME biobank patients included in the HEAL2 rare-variant study cannot be excluded, although not documented. The cell-type expression thresholds used to define foreground gene sets for ORA replication, while documented in the literature, remain arbitrary. Cell-type analysis was performed only on brain cells. Finally, this study focused on European subjects, and whether these results generalise to other ancestries is unknown.

5. Conclusions

In conclusion, our meta-GWAS and post-GWAS analyses of 19,470 ME/CFS cases provide evidence for the involvement of brain tissues, synapses, and specific neuronal cell types in the genetic architecture of ME/CFS. The independent replication of these findings using a gene module derived from rare variant analysis strengthens the validity of our results. Larger GWAS with greater ancestry diversity, combined with rare variant studies in clinically well-characterised cohorts, are needed to confirm and expand these findings and to further elucidate the pathophysiology of this debilitating disease.

Supplementary Materials: S1 File: Supplementary tables. The workbook contains a README sheet with full column descriptions and six data sheets: S1 (genomic risk loci, DME-1 + MVP meta-analysis), S2 (genomic risk loci, DME-1 alone), S3 (MAGMA gene-set enrichment analysis, MSigDB 2023.1.Hs), S4 (MAGMA gene-tissue expression analysis, GTEx v8), S5 (FUMA cell-type analysis, Siletti/Seeker human brain atlases), S6 (FUMA cell-type analysis, DropViz mouse brain atlas). All sheets include per-cohort statistics (DME-1 and MVP) and Zhang et al. replication results.

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Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ME/CFS	Myalgic encephalomyelitis/chronic fatigue syndrome
GWAS	Genome-wide association study
WGS	Whole-genome sequencing
MVP	Million Veteran Program
DME-1	DecodeME cohort 1
MAGMA	Multi-marker Analysis of GenoMic Annotation
FUMA	Functional Mapping and Annotation of GWAS
GTEx	Genotype-Tissue Expression project
ORA	Over-representation analysis
BH	Benjamini-Hochberg
eMSN	Eccentric medium spiny neuron
MSN	Medium spiny neuron
SEG	Specifically expressed gene
TDEP	Top decile expression proportion
LD	Linkage disequilibrium
MAF	Minor allele frequency
QC	Quality control
GWS	Genome-wide significant
SNP	Single nucleotide polymorphism
EHR	Electronic health record
IWMN	Interstitial white matter neuron

Appendix A. Mouse Brain Cell-Type Analysis

Table A1. Independent cell types from step 2 of the FUMA cell-type pipeline (DropViz mouse brain atlases; level-2 resolution; BH correction), with per-cohort statistics and HEAL2 module replication.

				DME-1 + MVP				DME-1			MVP			Replication	
Region	Cell type[a]	Class marker	Type marker	β	SE	P	P_{BH}	β	SE	P	β	SE	P	FE	P
STR	dSPN	Gad1Gad2	Drd1-Cxcl14	0.038	0.007	$\mathbf{3.96 \times 10^{-8}}$	2.24×10^{-5}	0.027	0.007	9.12×10^{-5}	0.007	0.007	1.68×10^{-1}	1.34	1.69×10^{-1}
GP	Th^+ SPN	Gad1Gad2-Th	Adora2a-Th	0.037	0.007	$\mathbf{1.76 \times 10^{-7}}$	5.00×10^{-5}	0.031	0.007	$\mathbf{1.34 \times 10^{-5}}$	−0.001	0.007	5.37×10^{-1}	1.24	2.61×10^{-1}
SN	Glut.	Slc17a6	Slc17a6.2	0.026	0.006	$\mathbf{1.54 \times 10^{-5}}$	7.90×10^{-4}	0.020	0.006	1.04×10^{-3}	0.005	0.006	2.07×10^{-1}	1.09	4.26×10^{-1}
CB	Granule	Slc17a7	Gabra6	0.024	0.006	$\mathbf{3.66 \times 10^{-5}}$	1.29×10^{-3}	0.022	0.006	3.14×10^{-4}	−0.003	0.006	6.71×10^{-1}	1.43	1.59×10^{-1}
PC	Layer6a	Slc17a7	Syt6-Sla	0.042	0.011	$\mathbf{3.89 \times 10^{-5}}$	1.29×10^{-3}	0.040	0.011	1.40×10^{-4}	−0.009	0.011	8.01×10^{-1}	0.86	7.05×10^{-1}
HC	DP	Slc17a7	C1ql2-Penk	0.034	0.009	$\mathbf{5.96 \times 10^{-5}}$	1.87×10^{-3}	0.034	0.009	$\mathbf{8.41 \times 10^{-5}}$	−0.007	0.009	7.79×10^{-1}	0.91	6.54×10^{-1}
SN	GABA.	Gad1Gad2	Gad2.3	0.030	0.008	1.00×10^{-4}	2.70×10^{-3}	0.030	0.008	2.04×10^{-4}	−0.000	0.008	5.15×10^{-1}	0.84	7.37×10^{-1}
HC	–	Slc17a7	Pvrl3-Grm5	0.025	0.007	1.36×10^{-4}	2.81×10^{-3}	0.020	0.007	2.29×10^{-3}	0.004	0.007	2.93×10^{-1}	1.58	4.90×10^{-2}
PC	Layer 5a	Slc17a7-Slc17a6	Rorb-Pcdh20	0.039	0.011	1.44×10^{-4}	2.81×10^{-3}	0.047	0.011	$\mathbf{1.14 \times 10^{-5}}$	−0.002	0.011	5.86×10^{-1}	1.55	9.74×10^{-2}
SN	VTA	Th	Cbln1	0.018	0.005	1.76×10^{-4}	3.11×10^{-3}	0.015	0.005	1.63×10^{-3}	0.007	0.005	7.80×10^{-2}	1.02	5.20×10^{-1}
FC	–	Gad1Gad2	Synpr-Pcdh11x	0.029	0.008	2.54×10^{-4}	4.10×10^{-3}	0.027	0.009	7.81×10^{-4}	0.000	0.008	4.84×10^{-1}	0.70	8.68×10^{-1}
FC	pyramidal	Slc17a7	Syt6-Kit	0.033	0.010	2.82×10^{-4}	4.30×10^{-3}	0.038	0.010	$\mathbf{6.40 \times 10^{-5}}$	−0.011	0.009	8.71×10^{-1}	1.29	2.47×10^{-1}
TH	NAN	Slc17a6	Rora-Cacng4	0.026	0.008	2.98×10^{-4}	4.44×10^{-3}	0.023	0.008	1.92×10^{-3}	0.002	0.008	3.74×10^{-1}	1.11	4.28×10^{-1}
CB	IC	Slc17a6	Nrgn	0.012	0.004	3.59×10^{-4}	4.57×10^{-3}	0.008	0.004	1.24×10^{-2}	0.001	0.004	4.14×10^{-1}	0.84	7.54×10^{-1}
TH	PAG	Gad1Gad2	Gata3	0.024	0.007	4.19×10^{-4}	4.73×10^{-3}	0.029	0.007	$\mathbf{4.28 \times 10^{-5}}$	0.002	0.007	3.62×10^{-1}	1.06	4.66×10^{-1}
HC	EC	Slc17a7	Nxph3-Sema3c	0.028	0.009	1.01×10^{-3}	8.61×10^{-3}	0.032	0.009	2.84×10^{-4}	−0.014	0.009	9.35×10^{-1}	0.99	5.62×10^{-1}
ENT	STN	Slc17a6	Pitx2	0.016	0.005	1.32×10^{-3}	1.05×10^{-2}	0.019	0.006	4.57×10^{-4}	0.000	0.005	4.72×10^{-1}	0.74	8.65×10^{-1}
PC	–	Gad1Gad2	Pvalb-Nefm	0.023	0.008	1.78×10^{-3}	1.26×10^{-2}	0.025	0.008	8.17×10^{-4}	0.008	0.008	1.56×10^{-1}	0.98	5.73×10^{-1}
HC	–	Gad1Gad2	Pvalb-Tac1	0.022	0.008	1.84×10^{-3}	1.29×10^{-2}	0.027	0.008	3.48×10^{-4}	0.012	0.008	5.30×10^{-2}	0.85	7.40×10^{-1}
GP	SI	Gad1Gad2	Lamp5-Cplx3	0.010	0.004	4.36×10^{-3}	2.34×10^{-2}	0.011	0.004	1.49×10^{-3}	0.000	0.004	4.69×10^{-1}	1.09	4.23×10^{-1}
PC	–	Gad1Gad2-Chat	Synpr-Slc5a7	0.020	0.008	4.87×10^{-3}	2.50×10^{-2}	0.018	0.008	1.28×10^{-2}	0.010	0.008	9.56×10^{-2}	1.85	9.03×10^{-3}

Twenty-one cell types passed Benjamini-Hochberg correction ($P_{BH} < 0.05$, $k = 565$) and FUMA step 2 conditional analysis in the DME-1 + MVP meta-analysis. Six of them also passed the more stringent Bonferroni correction and are indicated by bold P values. Bonferroni thresholds based on the number of cell types actually tested by MAGMA in each cohort: $P < 0.05/565 = 8.85 \times 10^{-5}$ for DME-1 + MVP, DME-1, and MVP ($k = 565$ for all three cohorts); $P < 0.05/564 = 8.87 \times 10^{-5}$ for genome-wide HEAL2 replication ($k = 564$ cell types with non-empty overlap with the HEAL2 background). Five of the 21 cell types survived Bonferroni correction in DME-1 alone (bold P values) and none reached nominal significance in MVP alone. Replication was performed as over-representation analysis of the HEAL2 module ($n = 115$ genes) [20] against the cell-type foreground gene set (17,759 HEAL2 background genes), implemented via the `enricher` function in `clusterProfiler` [42]. Two of the 21 cell types reached nominal significance for replication (a glutamatergic neuron in hippocampus and a GABAergic neuron in posterior cortex) but none survived Bonferroni correction for replication ($P < 0.05/21 = 2.38 \times 10^{-3}$). The complete results of cell-type analysis against the DropViz mouse brain atlas for all cohorts and replication are presented in S1 File, sheet S6. FC: frontal cortex; STR: striatum; GP: globus pallidus; ENT: entopeduncular nucleus; TH: thalamus; SN: substantia nigra; HC: hippocampus; PC: posterior cortex; CB: cerebellum. GABA.: GABAergic neuron; Glut.: glutamatergic neuron; dSPN: direct-pathway spiny projection neuron; DP: dentate principal cells; NAN: non-anterior nuclei; IC: inferior colliculus; PAG: periaqueductal gray; STN: subthalamic nucleus; SI: substantia innominata; Th^+ SPN: tyrosine hydroxylase-positive spiny projection neuron. We followed DropViz nomenclature [39] where Class Marker defines the broad neurotransmitter identity (e.g. *Gad1Gad2*: GABAergic; *Gad1Gad2-Th*: GABAergic with tyrosine hydroxylase co-expression; *Slc17a6*/VGluT2 and *Slc17a7*/VGluT1: glutamatergic) and Type Marker defines the specific subtype within that class. "Cell type" gives the curator-assigned label from the DropViz web portal [39]. β: MAGMA regression coefficient; SE: standard error; P: raw p-value; P_{BH}: Benjamini-Hochberg corrected p-value ($k = 565$); FE: fold enrichment. [a] All cell types in this table are neurons.

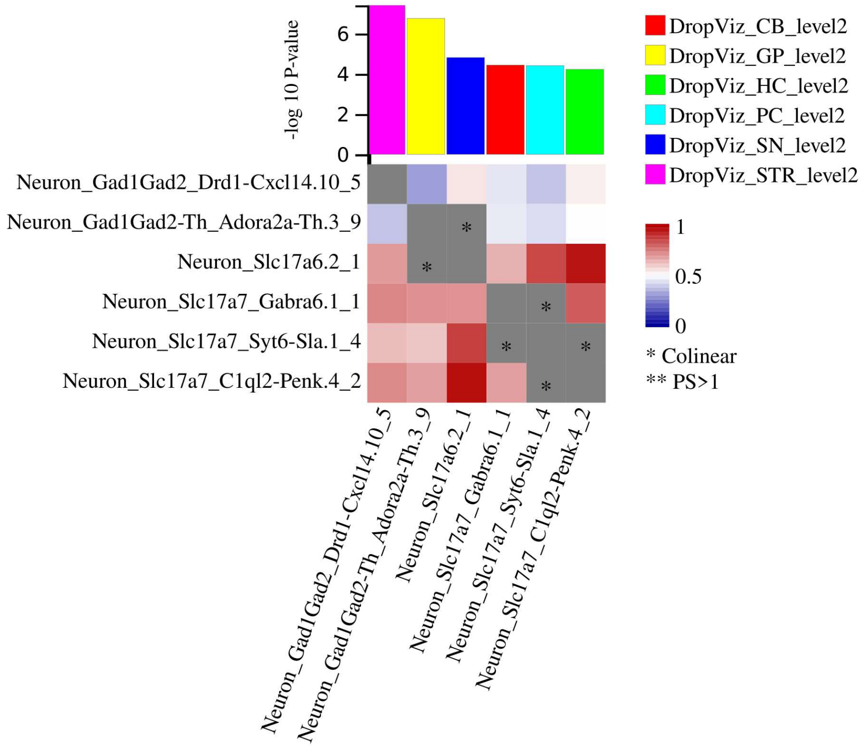

Figure A1. Step 3 conditional analysis from the FUMA cell-type pipeline (DropViz mouse brain atlas; level-2 resolution; Bonferroni correction; DME-1 + MVP meta-analysis). Cell types surviving Bonferroni correction in discovery and FUMA step 2 conditional analysis (Table A1) were subjected to cross-dataset conditional analysis for the detection of collinearity (FUMA step 3; see Figure 4 caption for interpretation of the heatmap). Three collinear pairs were identified (starred cells): the Th$^+$ SPN from GP and the glutamatergic neuron from SN; the granule cell from CB and the layer 6a neuron from PC; and the layer 6a neuron from PC and the dentate principal cell from HC.

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